Man-made green: An analysis of urban planning and sustainability in Medellín (Colombia)

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Abstract

This study measures the impact of the Siembra Aburrá urban planning and greening initiative in the city of Medellín, Antioquia, Colombia. The goal of this project was to increase vegetation in the city, reduce urban heat islands, and create more green spaces for the community. Using remotely sensed Sentinel-2 data, I assess vegetation increase via Normalized Difference Vegetation Index (NDVI) before (2015-2016), during (2016-2020) and after (2020-2021), the initiatives' active years. By analyzing the NDVI trends in the city's neighborhoods, I can evaluate whether the project's goals were met. Results show a significant increase in the year following the start of the project (2017-2018), but a dip the year after that (2018-2019) which might be associated with El Niño drought. However, while NDVI did increase, limitations of using NDVI in urban areas complicate the interpretation of results. This suggests the need for long term monitoring of the city in order to assess its sustainability and effectiveness. Further research could also integrate temperature and demographic data to get deeper into the broader impacts of vegetation increase on urban environments.

Keywords: *urban planning, green spaces, eco corridors, NDVI, satellite, vegetation, Medellín*

Introduction

Spatial resolution remotely sensed data of vegetation and urban environments offers the opportunity to evaluate the success of significant urban planning projects such as the Siembra Aburrá greening initiative set by the city of Medellín. The goal of the project was to increase vegetation and green spaces in the city, and the Ministry of the Environment of Medellín states to have planted over 1.2 million trees in the city since the beginning of the project. However, the city has lacked providing specific results and statistics on the project. Results provided do not clearly divide the results for each year, and fail to provide clear data showcasing an overall increase in vegetation in the city. Through remotely sensed data and GIS analysis on ESRI's ArcPro, I find relationships between the goals of the project and how it has impacted the urban areas of Medellín.

Principles

The main principle used in this project is the Normalized Difference Vegetation Index (NDVI). NDVI is calculated based on the fact that healthy vegetation reflects significant NIR and absorbs red light. More specifically, using this formula: (NDVI=(NIR-Red)/ (NIR+Red)\). Here, NIR is the reflectance in the NIR band (≈ 0.8 µm), and RED is the reflectance in the red band (≈ 0.6 µm). Results range from -1 to +1, where values closer to +1 indicate healthy and dense vegetation. Values close to 0 can indicate bare soil or built environments, and values close to -1 show surfaces that absorb at many wavelengths such as water.

Sentinel-2 data has a spatial resolution of 10 meters in the bands used for the NDVI calculation (visible and NIR bands), providing high-quality data for monitoring urban vegetation compared to a lower spatial resolution satellite such as Landsat (30 by 30 m). Additionally, Sentinel-2 has a temporal resolution of about 5 days. This short amount of time allows for frequent updates on the change in vegetation.

However, satellite imagery and therefore NDVI is sensitive to absorbed and scattered light, caused by environmental factors such as clouds, aerosols, and water vapor. Medellín especially is prone to cloudy days, an issue often encountered when producing NDVI calculations. Additionally, due to the topography and location of the city, pollution often stays trapped and forms an overcast over the areas we hope to analyze. Sentinel-2 is designed to reduce atmospheric interference in its imagery, but inaccuracies can still occur. Finally, the last consideration when using NDVI is that this calculation works best for relatively uniform vegetation types such as forests or grasslands. In complex urban environments, such as cities, vegetation may be fragmented or placed alongside non-vegetated areas such as buildings and roads. Nevertheless, NDVI is a way of measuring vegetation cover in cities comparable to classification methods outlined in other studies such as (Paniagua-Villada, et al., 2024).

Study area

The city of Medellín is located between the mountains of the Colombian Central Andes, in the northwestern part of South America. The city is at an elevation of 1,460 meters above sea level and covers around 16,030 hectares around the Aburrá river. Currently, Medellín is

Colombia's second largest city and had a population of 2,427,129 people recorded in 2018 (DANE Departamento Administrativo Nacional de Estadística). While the Siembra Aburrá project spans the entire 10 municipalities of the metropolitan area of Medellín, I decided to focus the study area on the most urbanized areas of the cities divided by the main neighborhoods to avoid the data being skewed by the highly rural areas right outside the city. Since my goal is to assess the difference between built cover and vegetation in the city itself, this area would provide redundant results shifting the results towards a higher mean NDVI.

Data and Methods

To source my satellite images, I used Sentinel 1C (Copernicus Service information) data provided by Google Earth Engine. I decided to use this over Landsat Imagery due to the higher spatial resolution and existing atmospheric correction. Initially, I researched the driest months annually in Medellín and found it to be between December and March. I also used a cloud mask to avoid cloudy scenes. However, such a short period of time and the current cloud mask used provided images covered by clouds. When NDVI was calculated using these images, they provided chunks of unavailable data when running analyses with Arc Pro. This required me to go back into the GEE code and apply a different cloud cover. I created Median NDVI Images from July 1st of each year, until June 30th of the following year from 2015-2021. This means I gathered 6 years of imagery, including 1 year before the project (2015-2016) and 1 year afterwards (2020-2021). I also gathered 4 years of data while the project was happening to be able to better understand trends as the project progressed.

The raster data for each year of median NDVI was exported to Arc Pro to conduct further analysis in the context of the specific neighborhoods of Medellín, using a shapefile of the neighborhoods contained by the most urbanized area of the city to be used as my bounds. First, I conducted a Zonal Statistics analysis to calculate the Mean NDVI of each neighborhood.

I then graphed the Mean NDVI of each year to get a better visualization of the trend the vegetation increase formed. I noticed that there was a peak in Mean NDVI a year after the project started (2017-2018) and a dip in 2018-2019. To validate results, I ran a t-test comparing the Mean NDVI of 2015/16 (start of the project) to 2017/18 (the year where NDVI peaked). To further prove the increase on NDVI, I ran more t-tests comparing this base year against each other. The relationship of higher NDVI was held constant. Specific results expanded upon in Appendix A.

To prove further significance I used the Reclassify raster tool on Arc Pro to conduct a reclassification of the raster through a conditional statement. The conditional statement used was the following: any pixel less than 0.2 in NDVI was classified as 0, and any pixel higher than 0.2 was classified as 1. Through this, I could see the increase of pixels classified as 'High NDVI' in the city. I then performed a zonal statistics analysis for each neighborhood and created a sum of pixels classified as 1 (above 0.2 in NDVI) for each year. I then took the area of the sum of each neighborhood and plotted each year's results on a graph. Results outlined in Figure 3. I then subtracted mean NDVI in 2015/16 to 2020/21 and ran another zonal statistics analysis to find the neighborhoods with the most increase in NDVI between the two years. This was also done subtracting the peak of the project 2017/18 from the start 2015/16. Additionally, I ran six t-tests using Microsoft Excel to determine the significance of the change in the sum of pixels throughout the six years compared to 2015-2016. Specific results expanded upon in Appendix B.

Results

Trends stay consistent between the data analysis of mean NDVI and the pixel reclassification. In both analyses, NDVI peaked in the time period of 2017-2018 and decreased significantly the following year. After that, NDVI didn't show dramatic change in both scenarios through 2019-2020 and 2020-2021. Results are outlined below.

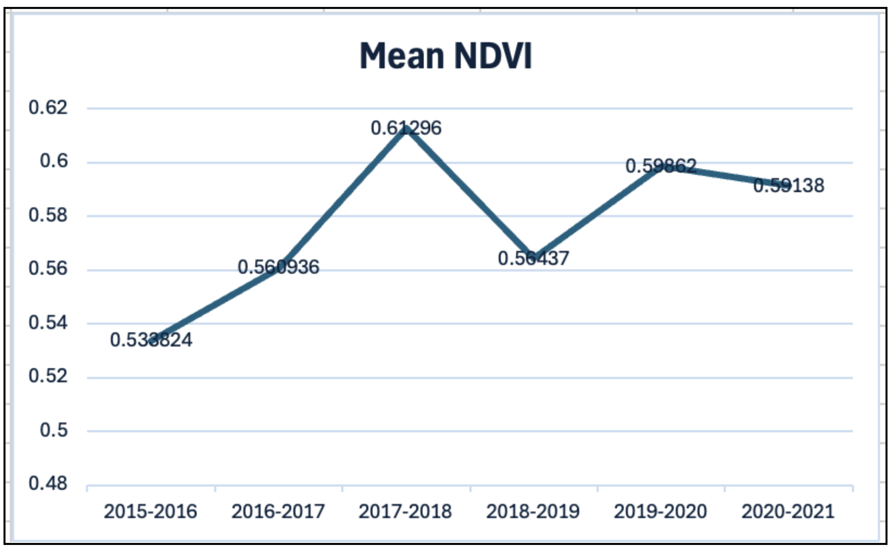

Figure 1. Mean NDVI in the city of Medellín from 2015 to 2021.

T-test results between the base year 2015-2016 and the peak in 2017-2018 yielded a p-value of 3.15 e^-95 < 0.05. We can conclude with almost 100% confidence that the year 2017-2018 was higher in mean NDVI than right before the project began. The dip in increasing

NDVI found in 2018-2019 still yielded a significant p-value despite leaning closer to 0.05 than any of the other years.

The neighborhoods with the highest increase in NDVI between 2015-2016 and 2020-2021 were Cerro el Volador (+0.143), El Remanso (+0.133), Cerro Nutibara (+0.123), Altos del Poblado (+0.105), El Rodeo (+0.099).

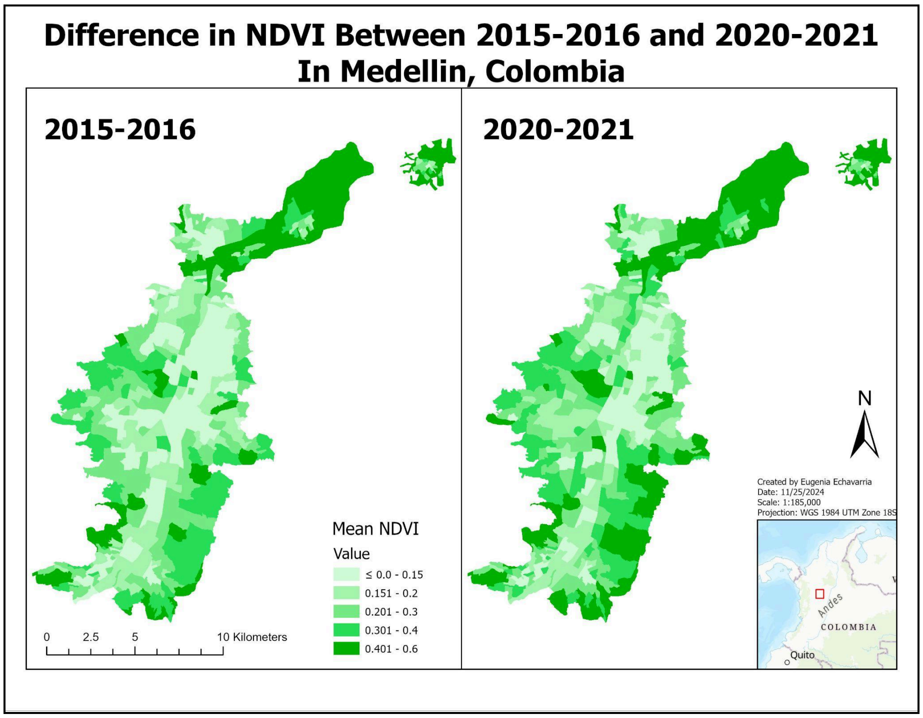

Figure 2. Comparison between the beginning (2015/17) and end of the project (2020/2021) based on NDVI Calculation processes in GEE from Sentinel-2 imagery.

The neighborhoods with the highest increase in mean NDVI between 2015-2016 and 2017-2018 were Cerro el Volador (+0.157), Cerro Nutibara (+0.118), El Remanso (+0.114), Hector Abad Gomez (+0.114) and Espiritu Santo (+0.105).

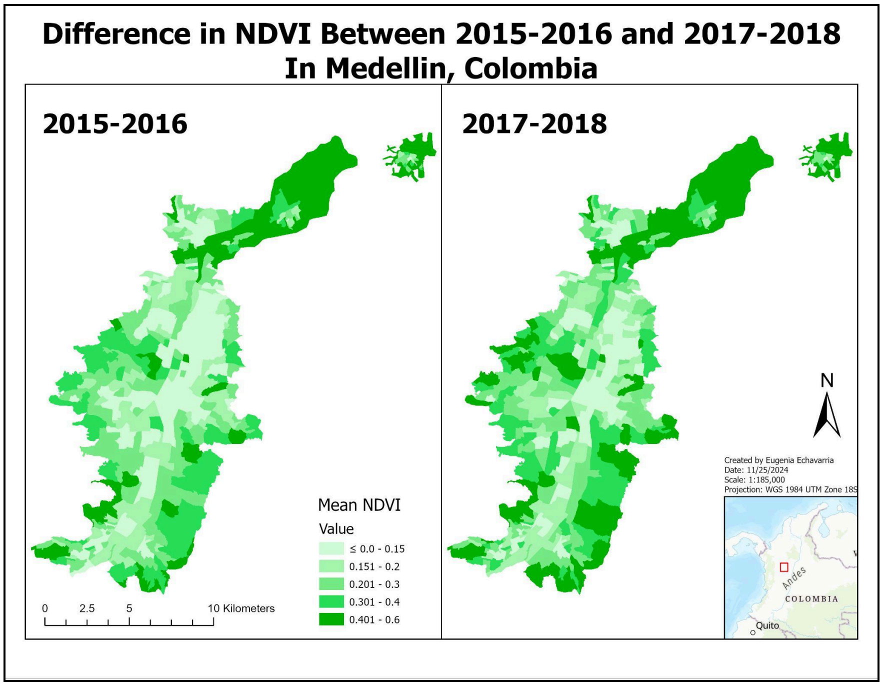

Figure 3. Comparison between the beginning (2015/17) and peak of the project (2017/2018) based on NDVI Calculation processes in GEE from Sentinel-2 imagery.

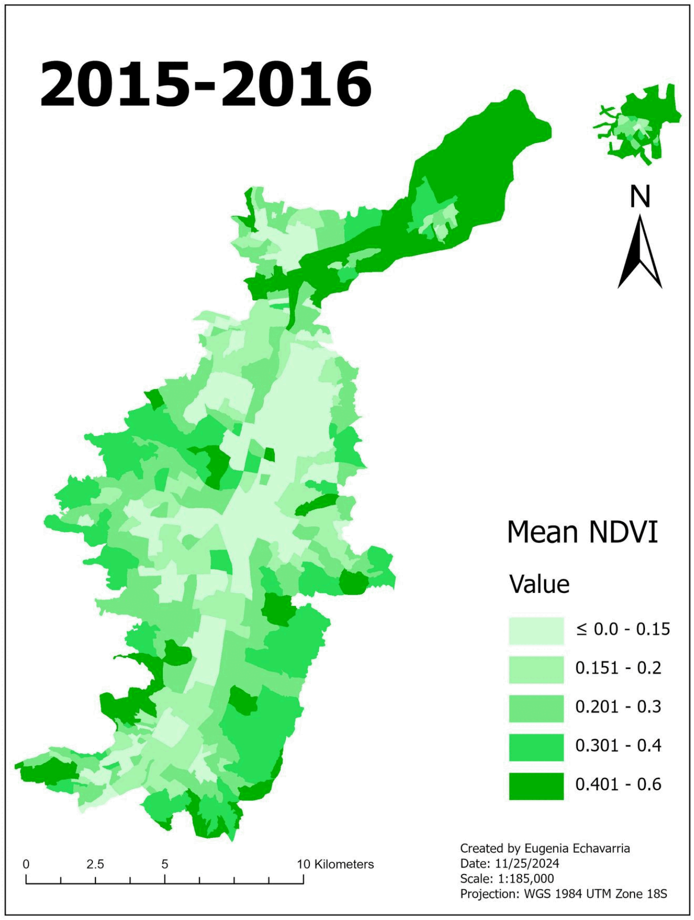

Figure 4. Mean NDVI between 2015 and 2021 in Medellin, Antioquia, Colombia.

The pixel classification method yielded similar results to Mean NDVI. The base year 2015-2016 proves to be the lowest between all six, 2017-2018 remains the peak, and the drop between 2017-2018 remains large.

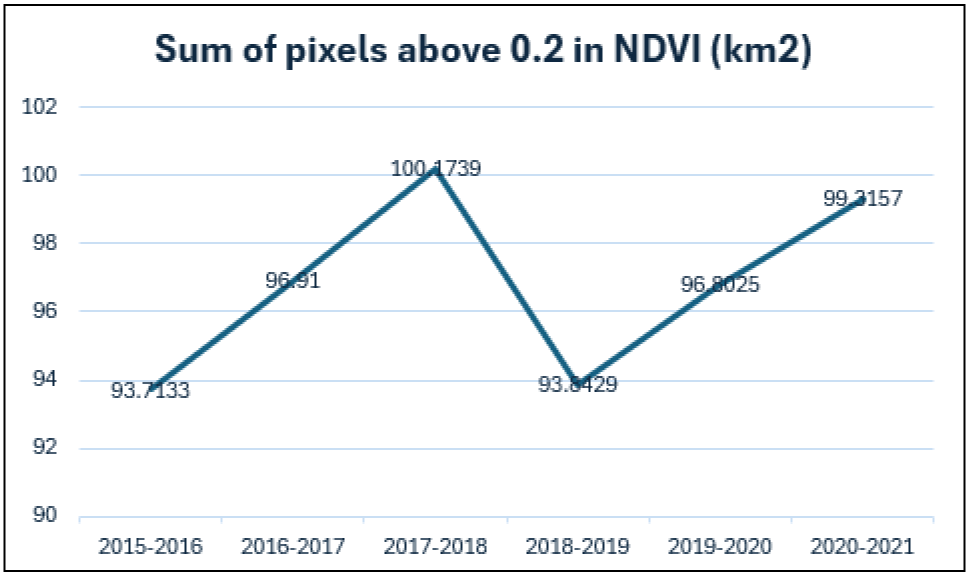

Figure 5. Total area of the sum of pixels classified above 0.2 NDVI in the city of Medellín.

T-test results conclude that the difference in the sum of pixels above 0.2 NDVI is significant for the peak in 2017-2018, which yielded a p-value of 4.69 e^-39 < 0.05. Like NDVI, we can also conclude with almost 100% certainty that the increase in sum of pixels above 0.2 NDVI is significant between 2015-2016 and 2017-2018. This relationship held true for all other years except 2018-2019, which proved to be a non-significant change compared to 2015-2016.

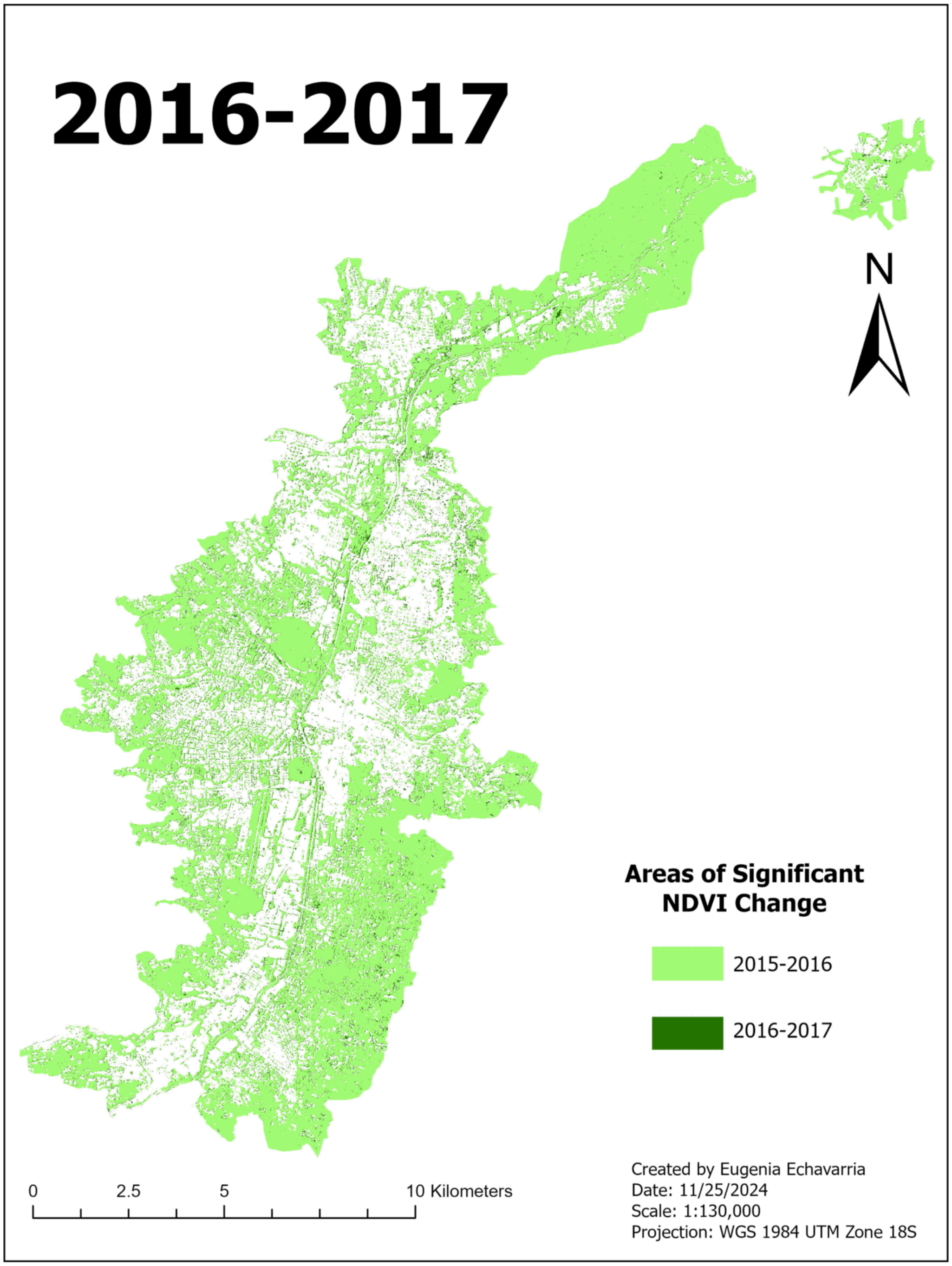

Figure 6. Change in area above 0.2 in NDVI comparing 2015/16 to every year after that until 2021.

The dip in NDVI for both data analysis methods proved to remain consistent. However, results line up with the onset of the El Niño phase in late 2018. By late 2018, the "IDEAM (Institute of Hydrology, Meteorology and Environmental Studies) in Colombia had already placed warnings for the then upcoming 2018–2019 drought phase and dry condition." (Weng, et al, 2020)

Conclusions

The green corridors project or Siembra Aburrá seemed to have proven effective at increasing the vegetation in the city of Medellín. Especially in the year following the start of the project, increase in vegetation is significant in many neighborhoods. However, NDVI makes it hard to differentiate different types of vegetation, compared to other methods such as classification (Paniagua-Villada, et al., 2024) making it difficult to assess the specific cause for the increase. It would be useful and interesting to be able to properly see whether vegetation is new, or just expanded from the already existing natural vegetation in the city. Additionally, this project should be analyzed long term to consider the time it takes for saplings to grow into mature trees. While the drought in 2018 could be an explanation for the dip in NDVI that year, I believe that the limitations on NDVI analysis in urban environments discussed earlier leaves a lot of room for error. Instead, the peaks could simply be growing vegetation in years following average weather conditions. Dips in NDVI could also be strongly correlated to other drought events. While I attempted to be able to find significant results, it is difficult to follow alongside the Siembra Aburrá project since it has such loosely outlined methods, goals and results. If I had three months to work on this project I would not only more carefully analyze specific areas to determine what type of vegetation increased in the city, but also compare the increase in vegetation to urban heat islands and census demographics. I would like to see if there is a connection between temperature and vegetation in Medellín, and what types of neighborhoods are the most and least affected. I believe measuring the success of a project should include the impact on people and the sustainability of the city itself rather than mere increase, but with the limited time for the project it could not be done.

Appendix A

T test results conducted in Microsoft Excel depicting p values for Mean NDVI between 2015/16 vs. 2016/17, 2015/16 vs. 2017/18, 2015/16 vs 2018/19, 2015/16 vs 2019/2020, and 2015/16 vs 2020/2021.

Mean NDVI

t-Test: Paired Two Sample for Means

	(15/16)	*(16/17)*
Mean	0.220556	0.239076
Variance	0.010917	0.012425
Observatio	448	448
Pearson Co	0.981147	
Hypothesiz	0	
df	447	
t Stat	-17.7453	
P(T<=t) one	5.08E-54	
t Critical on	1.64827	
P(T<=t) two	1.02E-53	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	*17/18*
Mean	0.220556	0.253018
Variance	0.010917	0.013451
Observatio	448	448
Pearson Co	0.978277	
Hypothesiz	0	
df	447	
t Stat	-26.7736	
P(T<=t) one	3.15E-95	
t Critical on	1.64827	
P(T<=t) two	6.3E-95	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	*18/19*
Mean	0.220556	0.226428
Variance	0.010917	0.011261
Observatio	448	448
Pearson Co	0.971433	
Hypothesiz	0	
df	447	
t Stat	-4.92758	
P(T<=t) one	5.87E-07	
t Critical on	1.64827	
P(T<=t) two	1.17E-06	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	*19/ 20*
Mean	0.220556	0.239291
Variance	0.010917	0.01173
Observatio	448	448
Pearson Co	0.976185	
Hypothesiz	0	
df	447	
t Stat	-16.8534	
P(T<=t) one	5.4E-50	
t Critical on	1.64827	
P(T<=t) two	1.08E-49	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	*20/21*
Mean	0.220556	0.248049
Variance	0.010917	0.012778
Observatio	448	448
Pearson Co	0.973937	
Hypothesiz	0	
df	447	
t Stat	-22.1721	
P(T<=t) one	2.57E-74	
t Critical on	1.64827	
P(T<=t) two	5.13E-74	
t Critical tw	1.965285	

Appendix B

T test results depicting p values for Sum of pixels above 0.2 in NDVI between 2015/16 vs. 2016/17, 2015/16 vs. 2017/18, 2015/16 vs 2018/19, 2015/16 vs 2019/2020, and 2015/16 vs 2020/2021.

Pixel

t-Test: Paired Two Sample for Means

	15/16	16/17
Mean	2091.815	2163.17
Variance	1.04E+08	1.06E+08
Observatio	448	448
Pearson Co	0.999898	
Hypothesiz	0	
df	447	
t Stat	-7.74939	
P(T<=t) one	3.13E-14	
t Critical on	1.64827	
P(T<=t) two	6.27E-14	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	17/18
Mean	2091.815	2236.025
Variance	1.04E+08	1.07E+08
Observatio	448	448
Pearson Co	0.999899	
Hypothesiz	0	
df	447	
t Stat	-13.6614	
P(T<=t) one	4.69E-36	
t Critical on	1.64827	
P(T<=t) two	9.39E-36	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	18/19
Mean	2091.815	2094.708
Variance	1.04E+08	1.02E+08
Observatio	448	448
Pearson Co	0.999879	
Hypothesiz	0	
df	447	
t Stat	-0.31849	
P(T<=t) one	0.375132	
t Critical on	1.64827	
P(T<=t) two	0.750265	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	19/20
Mean	2091.815	2160.77
Variance	1.04E+08	1.01E+08
Observatio	448	448
Pearson Co	0.999859	
Hypothesiz	0	
df	447	
t Stat	-6.81036	
P(T<=t) one	1.58E-11	
t Critical on	1.64827	
P(T<=t) two	3.15E-11	
t Critical tw	1.965285	

t-Test: Paired Two Sample for Means

	15/16	20/21
Mean	2091.815	2216.868
Variance	1.04E+08	1.05E+08
Observatio	448	448
Pearson Co	0.999827	
Hypothesiz	0	
df	447	
t Stat	-12.8768	
P(T<=t) one	8.57E-33	
t Critical on	1.64827	
P(T<=t) two	1.71E-32	
t Critical tw	1.965285	

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